

Key Concepts

- **Key Concepts:**
 - End-to-end Requirements Traceability Matrix (RTM),
 - defining SLA/NFR criteria (security, load, latency, concurrency),
 - UAT sign-off documentation.

1. End-to-End Requirements Traceability Matrix (RTM)

What is an RTM?

A **Requirements Traceability Matrix (RTM)** is a document that tracks every requirement throughout the entire project lifecycle to ensure that:

- Every requirement is implemented
- Every requirement is tested
- Nothing is missed
- Changes can be traced back to the original business need

Think of it as a **mapping table** that connects:

Business Need → Requirement → Design → Development → Test Case → UAT → Production

This is why it is called **end-to-end traceability**.

Why is RTM important?

Without an RTM:

- Requirements may be forgotten
- Developers may build the wrong feature
- Testers may miss testing something
- Stakeholders cannot verify coverage

An RTM helps answer:

- Did we build everything requested?
- Did we test everything built?
- Which test cases validate a requirement?
- Which requirements are impacted by a change?

Types of Traceability

Forward Traceability

Requirement → Test

Example:

BR-001 → TC-201

Ensures **every requirement** gets tested.

Backward Traceability

Test → Requirement

Example:

TC-201 → BR-001

Ensures **no unnecessary** functionality is built.

Bidirectional Traceability

Both directions.

This is considered best practice.

2. Defining SLA and NFR Criteria

These are often confused but serve different purposes.

SLA (Service Level Agreement)

An SLA defines:

The measurable level of service that must be delivered.

Usually agreed between:

- Business and IT
- Customer and Vendor

Examples

Metric	SLA
Availability	99.9% uptime
Response Time	< 2 seconds
Incident Response	Within 30 minutes
Resolution Time	Within 4 hours

NFR (Non-Functional Requirements)

NFRs define:

How well the system should perform.

Functional Requirement:

User can submit an order.

NFR:

Order submission must complete within 2 seconds.

Common NFR Categories

1. Security

Defines protection requirements.

Examples:

- MFA required
- Password minimum 12 characters
- Data encrypted at rest
- Data encrypted in transit
- Role-based access control

Example Security NFR

All customer data must be encrypted using AES-256.

2. Load

Load refers to:

How much work the system must handle.

Examples:

- 5,000 users/day
- 50,000 transactions/hour
- 1 million API calls/day

Example

An online university system:

- 20,000 students
- 2,000 concurrent logins during enrollment

Load testing validates this.

3. Latency

Latency means:

Time taken to respond.

Examples:

- API response < 500 ms
- Dashboard load < 3 seconds

Example

Example

Poor latency:

User clicks "Submit"

Waits 8 seconds

Good latency:

User clicks "Submit"

Response in 1 second

4. Concurrency

Concurrency means:

Number of users operating simultaneously.

This differs from total users.

Example:

- 100,000 registered users
- 2,000 active at same time

Concurrency is 2,000.

Example NFR

The system shall support 3,000 concurrent users without performance degradation.

Sample NFR Table

Category	Requirement
Security	AES-256 encryption
Availability	99.95% uptime
Load	10,000 transactions/hour
Latency	API response < 500ms
Concurrency	3,000 users simultaneously
Scalability	Scale to 5x current load
Reliability	Recovery within 15 minutes
Backup	Daily backup retention 30 days

3. UAT Sign-Off Documentation

What is UAT?

User Acceptance Testing (UAT) is the final validation performed by business users before a solution goes live.

Goal:

Confirm the system meets business requirements.

Unlike system testing:

- Developers verify technical correctness.
- Business users verify business suitability.

UAT Process

Step 1

Prepare UAT scenarios.

Example:

- Create customer
- Generate invoice
- Reset password

Step 2

Business users execute scenarios.

Step 3

Record results.

Scenario	Result
Create Customer	Pass
Generate Invoice	Pass
Password Reset	Fail

Step 4

Fix defects.

Step 5

Retest.

Step 6

Obtain sign-off.

What is UAT Sign-Off?

A formal document confirming:

- Testing completed
- Requirements satisfied
- Business accepts solution
- Go-live approved

Typical UAT Sign-Off Template

Project

Student Management System Upgrade

UAT Summary

Item	Result
Total Scenarios	45
Passed	43
Failed	0
Deferred	2
Critical Defects	0

Acceptance Statement

The business confirms that all critical requirements have been successfully tested and accepted. The solution is approved for production deployment.

Signatories

Name	Role	Signature
Business Sponsor	Director	Signed
Product Owner	Product Owner	Signed
BA Lead	Business Analyst	Signed
Project Manager	PM	Signed

Why UAT Sign-Off Matters

Without sign-off:

- No evidence of business acceptance
- Risk of disputes after go-live
- Difficult accountability

With sign-off:

with sign-off:

- Formal approval exists
- Go-live decision documented
- Project closure supported

How These Three Concepts Connect

A mature project delivery flow looks like this:

Business Requirements



Requirements Traceability Matrix (RTM)



Functional Requirements



NFR / SLA Definition



Development



System Testing



UAT Testing



UAT Sign-Off



Production Release